

Mapping (Version 1.0)

Data confidentiality and integrity SHALL use the AES-CCM mode as defined in [NIST 800-38C](#) with the following parameters:

- `int CRYPTO_SYMMETRIC_KEY_LENGTH_BITS := 128` (this is the key length of the underlying block cipher in bits)
- `int CRYPTO_SYMMETRIC_KEY_LENGTH_BYTES := 16` (this is the key length of the underlying block cipher in bytes)
- `int CRYPTO_AEAD_MIC_LENGTH_BITS := 128` (this is the MIC length in bits)
- `int CRYPTO_AEAD_MIC_LENGTH_BYTES := 16` (this is the MIC length in bytes)
- `int CRYPTO_AEAD_NONCE_LENGTH_BYTES := 13`
- Key length SHALL be `CRYPTO_SYMMETRIC_KEY_LENGTH_BITS` bits.
- MIC length SHALL be `CRYPTO_AEAD_MIC_LENGTH_BITS` bits.
- The parameter `q` SHALL be 2 (length of encoding of maximum length) as specified in Appendix A.1 of [NIST 800-38C](#).
- The parameter `n` SHALL be `CRYPTO_AEAD_NONCE_LENGTH_BYTES` (length of nonce in bytes) as specified in Appendix A.1 of [NIST 800-38C](#).

`SymmetricKey` is an opaque data type to hold a symmetric block cipher key or any handle or reference that allows other primitives to access the corresponding key.

3.6.1. Generate and encrypt

Function and description

```
byte[lengthInBytes(P) + CRYPTO_AEAD_MIC_LENGTH_BYTES]
Crypto_AEAD_GenerateEncrypt(
    SymmetricKey K,
    byte[lengthInBytes(P)] P,
    byte[] A,
    byte[CRYPTO_AEAD_NONCE_LENGTH_BYTES] N)
```

Performs the generate and encrypt computation on payload `P` and the associated data `A` using the key `K` and a nonce `N`; the output contains the ciphertext and the tag truncated to `Tlen` bits (the encoding of the output depends on the mapping to the specific instance of the cryptographic primitive).